Physics from Existence I: The Equation

The Binary-Entropy Equation $V = -H$ and the Structure of Standard-Model Parameters

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Abstract

No theory has simultaneously derived all 26 free parameters of the Standard Model. In this paper, we show that three axioms about existence—existence is bivalent ($n \in \{0,1\}$, i.e. $n^2{=}n$), existence is uncertain ($\sigma{\neq}1$), and non-existence is forbidden ($\sigma{\neq}0$)—uniquely determine the single equation $V = -H$ as an algebraic identity. The mathematical structure generated by this equation corresponds to that of the Standard Model, reproducing more than 20 physical quantities with no adjustable parameters, all agreeing with experiment to 0.0006%–2.6% accuracy.

$V = -H$ is the canonical free energy of a single binary degree of freedom $n \in \{0,1\}$, determined as the Legendre transform of the Fermi–Dirac distribution. The Schrödinger equation is derived as a transfer matrix from this potential and the kinetic term of the canonical normalisation ($g = 1$, §2.4). This equation has exactly three bound states (discrete solutions), corresponding to the three generations of fermions. The spatial dimension $d = 3$ follows from three conditions—preservation of the well, spin-statistics, and propagating gravity. The mathematical structure unfolding from this coincidence reproduces the gauge group SU(3)×SU(2)×U(1).

More than 20 physical constants are derived from this structure, including $1/\alpha = 137.0368$ (0.0006%), $\alpha_s = 0.1179$ (0.01%), $\sin^2\theta_W = 3/13$ (0.2%), the charged-lepton mass ratio $R = 1.5293$ (0.006%), and $\sin\theta_C = 0.2245$ (0.10%). The results further predict the absolute neutrino masses ($\Sigma m_\nu = 59.2$ meV) and strictly zero neutrinoless double-beta decay, yielding seven testable predictions. These quantities are not independent input parameters—they are all computed from the same mathematical structure fixed by $V = -H$.

1 Introduction

The Standard Model is the most successful theory of particle physics. It describes all known particle interactions and has withstood extensive experimental tests, including the discovery of the Higgs boson [1, 2] (2012). Yet the theory contains 26 free parameters[1]—fermion masses (Yukawa couplings), mixing angles, gauge couplings, the Higgs

[1] The minimal Standard Model with massless neutrinos has 19 free parameters [22]; adding three neutrino masses and four PMNS parameters gives 26. Some references count 25, absorbing v_{EW} into the energy scale.

self-coupling, and CP phases. Their values must be measured and input by hand; why they take those values is not explained within the Standard Model. The generation number $N = 3$ is likewise an unexplained input. Despite many attempts—grand unified theories (GUTs), supersymmetry (SUSY), string theory—no theory has simultaneously determined all 26 parameters with zero free parameters [3]. Even a single constant, the fine-structure constant α, has a long history of attempts at first-principles derivation [35, 36, 37], but these have remained isolated numerical coincidences, lacking a unifying equation or mechanism.

An alternative approach seeks to derive physics from information theory. Wheeler's "it from bit" [4] proposed that physical reality arises from information but lacked a mathematical framework. Jaynes [5] derived statistical mechanics from entropy maximization but did not reach quantum field theory. Verlinde [7] derived gravitational force from entropic reasoning but addressed gravity alone. Other approaches include Frieden's [6] Fisher information, Caticha's [8] information geometry, and the informational axiomatisations of Hardy [9] and Chiribella et al. [10]. All of these approaches share a common limitation: they derive the *form* of physical equations but not the *values* of the constants that appear in them. The closest published programme in spirit is the spectral Standard Model of Connes and Chamseddine [39, 40, 41], which derives the SM gauge group, Higgs sector, and parameter relations from a noncommutative-geometric structure; its parameter-free prediction $m_H \approx 170$ GeV was falsified and later repaired by adding a singlet field—a direct cautionary precedent for parameter-free mass predictions, to which the predictions of this paper are equally subject. A related algebraic reconstruction is Furey's division-algebra approach [42].

This paper starts from three axioms: Existence ($n \in \{0, 1\}$), Uncertainty ($\sigma \neq 1$), and Non-existence is forbidden ($\sigma \neq 0$). Together A2 and A3 determine $\sigma \in (0, 1)$, a partition function $Z = 1 + e^{\phi}$, and, via the Legendre transform of the Fermi–Dirac distribution, the canonical free energy $V = -H(\sigma(\phi))$ as an algebraic identity, where H is the binary Shannon entropy, $\sigma(\phi)$ is the Fermi–Dirac distribution (sigmoid function), and $\phi = \mathrm{logit}(\sigma)$ is the natural parameter [14, 15]. The Schrödinger equation is derived as a transfer matrix from this potential and the kinetic term of the canonical normalisation ($g = 1$, §2.4).

This Schrödinger equation has exactly three bound states ($N = 3$), corresponding to the three generations of fermions. The spatial dimension $d_{\text{space}} = 3$ follows from three conditions—preservation of the well, spin-statistics, and propagating gravity; the spacetime dimension $D = 4$ follows from gravitational degree-of-freedom matching and $\mathbb{C}P^2$, and the signature $(3, 1)$ follows as a connection to the $M_2(\mathbb{C})_{sa}$ carrier structure (§9). The gauge structure $\mathrm{PG}(2, \mathbb{F}_3)$ emerges from the coincidence $N = d = 3$, corresponding uniquely to the gauge group $\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)$. From the representations of this gauge group, a structure yielding 48 fermions, 12 gauge bosons, and 1 Higgs is derived (§5). More than 20 parameters are derived—including $1/\alpha_{\text{em}} = 137.0368$ (0.0006%), $\sin^2\theta_W = 3/13$ (0.2%), the charged-lepton mass ratio $R = 1.5293$ (0.006%), Koide's $Q \approx 3/2$ ($\delta = -1.5 \times 10^{-5}$), CKM and PMNS mixing angles, and the bare $\theta_{\text{QCD}} = 0$ structure—all at 0.0006%–2.6% accuracy with zero adjustable parameters (§10).

Beyond the numerical agreement, the mathematical structure derived from $V = -H$ corresponds to structural problems of the Standard Model. The role played in the Standard Model by $\mu^2 > 0$—moving the vacuum from the symmetric point to a finite field value—is played here by the shape of the well of V (§7); the bounded internal potential ($V \in [-\ln 2, 0]$) supplies no instability channel, so neither the metastability problem nor the hierarchy problem arises within this construction (the correspondence to 4D vacuum stability and the hierarchy problem requires the reconstruction matching). Since $V = -H$ is uniquely determined as the canonical free energy of $n \in \{0, 1\}$, the Higgs is fixed to be unique.

This paper shows that, when the mathematical structure generated by $V = -H$ is placed in correspondence with the structure of the Standard Model, the resulting numerical values agree with experiment; it does not claim to derive the Standard Model itself rigorously from first principles. The refinement and uniqueness of this correspondence, together with the operator-level derivations of the individual constructions in this paper, are treated in Paper II [20].

The paper is organized as follows. Section 2 derives $V = -H$ from three axioms and establishes the Schrödinger equation. Section 3 proves the bound-state count $N = 3$ as a mathematical theorem. Section 4 derives the spatial dimension $d = 3$ as the unique common solution of three independent physical requirements. Section 5 constructs the gauge structure from $\mathrm{PG}(2, \mathbb{F}_3)$. Section 6 derives the gauge coupling constants. Section 7 derives the Higgs mass and shows that the vacuum is pinned at an interior point with the internal potential bounded (no instability channel). Section 8 places the bound states in correspondence with the three generations of charged leptons and derives fermion masses and mixing. Section 9 confirms the spacetime dimension $D = 4$ and signature $(3, 1)$ by consistency checks with stated premises, and establishes the tensor-product structure $\mathcal{H}_{\text{int}}(\phi) \otimes \mathcal{H}_{\text{space}}(x)$ from the independence of the construction. Section 10 summarises all results and presents falsifiable predictions. Section 11 gives the conclusion.

2 From Existence to the Equation

This chapter formalises three axioms about existence mathematically and derives the fundamental equation $V = -H$ through an algebraic transformation. From this equation, a structure consistent with the Standard Model emerges.

2.1 The Axioms

We define the three axioms on which the entire structure of this paper rests.
A1 (Existence): Something exists. "Exists" is a proposition; it is either true or false — by the law of excluded middle, there is no third value. Writing the state of existence as n, the state space is $n \in \{0, 1\}$. A1 defines the state space but does not determine the **value** of n.

A2 (Uncertainty): Existence is uncertain. Whatever exists does not exist with certainty: the occupation probability $\sigma = \langle n \rangle$ satisfies $\sigma \neq 1$. A2 excludes the upper endpoint (certain existence) but says nothing about the lower endpoint.

A3 (Prohibition of non-existence): Non-existence is not a state ($\sigma \neq 0$). A way of being that consists in not-being is a contradiction.

From these three axioms, the structure and constants of the Standard Model follow. We now formalise them.

2.2 Bivalence and the Partition Function

Starting from the state space $n \in \{0, 1\}$ established by A1, we construct the partition function.

Since n takes only the values 0 and 1,

$$n^2 = n \tag{1}$$

follows — because the solutions of $n(n-1) = 0$ are exactly $\{0, 1\}$. However, A1 alone leaves the system in a pure state $n = 0$ or $n = 1$, with no dynamics. A2 ($\sigma \neq 1$) excludes certain existence; A3 ($\sigma \neq 0$) excludes non-existence. Together they determine $\sigma \in (0, 1)$: the system is described as a canonical ensemble of a binary variable, neither certainly present nor certainly absent. $n^2 = n$ truncates the sum to two terms, and the partition function is

$$Z = \sum_{n=0}^{1} e^{n\phi} = 1 + e^{\phi}, \tag{2}$$

where $\phi = \ln[\sigma/(1-\sigma)]$ (the logit transform) is the canonical coordinate of the exponential family. Were n able to take values $0, 1, 2, \ldots$ (Bose statistics), the sum would be an infinite series. As a consequence of the law of excluded middle, $n^2 = n$ forbids this and restricts the occupation number to $\{0, 1\}$—the Fermi–Dirac occupation statistics. Exchange antisymmetry (spin statistics) does not follow from this alone; it is carried by the many-body realisation (§5.2).

The expectation $\sigma = \langle n \rangle = \partial \ln Z/\partial\phi$ coincides with the Fermi–Dirac distribution, and its variance is the source of all subsequent structure:

$$\sigma = \frac{1}{1 + e^{-\phi}}, \qquad g_F = \sigma(1-\sigma) = \mathrm{Var}(n) > 0. \tag{3}$$

These are not assumed but derived, from the combination of A1 (bivalence), A2 (uncertainty: $\sigma \neq 1$), and A3 (non-existence forbidden: $\sigma \neq 0$). g_F is the gap between the binary observable ($n^2 = n$) and its uncertain expectation ($\sigma^2 \neq \sigma$), and vanishes only at the pure values $\sigma \in \{0, 1\}$.

2.3 The Effective Potential $V = -H$

§2.2 established the partition function $Z = 1 + e^{\phi}$. The next step is to construct an effective potential from it.

The naive choice $-\ln Z$ diverges as $\phi \to \infty$, so a Legendre transform to the canonical free energy is required. The logarithm $W = \ln Z$ of the partition function generates the classical field $\sigma = dW/d\phi$ and yields the effective potential:

$$V(\phi) = \phi\sigma - W(\phi) = \sigma \ln \sigma + (1-\sigma) \ln(1-\sigma) = -H(\sigma), \tag{4}$$

where $H(p) = -p \ln p - (1-p) \ln(1-p)$ is binary Shannon entropy [12].

That is, $V = -H$ is an algebraic identity: the canonical free energy of a single binary degree of freedom $n \in \{0, 1\}$ equals the negative of the Shannon entropy. There is no freedom of choice. The potential

$$V = -H \tag{5}$$

is thereby uniquely determined by axioms A1–A3. The only input is three axioms about existence; no physics is assumed. From this single equation, a structure closely matching the Standard Model emerges, and values agreeing with physical constants to high precision are derived (§3–§10).

2.4 The Schrödinger Equation

§2.3 established $V = -H$ as the potential. The next question is how many discrete states this potential well can hold. To answer it, an eigenvalue equation is needed, which requires a kinetic term in addition to the potential. The kinetic term likewise follows from the same structure (its relative normalisation is fixed by $g = 1$ below).

The exponential family of a binary variable has two natural flat coordinates: $u = 2\arcsin\sqrt{\sigma}$, in which the Fisher metric is flat, and $\phi = \mathrm{logit}(\sigma)$, in which the e-connection is flat [13, 14, 15]. A3 forbids $\sigma = 0$. The coordinate u places $\sigma = 0$ at the finite point $u = 0$, so defining the dynamics requires boundary-condition data at that point—the point of non-existence becomes a repository of model-defining data. The coordinate ϕ sends $\sigma = 0$ to $\phi \to -\infty$ and requires no boundary data. A3 therefore selects ϕ. Free motion on the e-flat coordinate ϕ has a Lagrangian that is a constant-coefficient quadratic form in $\dot{\phi}$.

The choice of ϕ fixes the coordinate, but not the relative normalisation of the kinetic term and the action weight—the remaining freedom is the single dimensionless ratio $g = b/a^2$. The scale of ϕ is fixed by the binary indicator of A1, the scale of $V = -H$ by the Legendre identity, and rescaling the step size changes a^2 and b in the same proportion, leaving g unchanged. This paper adopts the canonical normalisation $g = 1$, which measures the quadratic variation per step of ϕ and the action weight as quantities of the same transfer process.[2]

These choices—the canonicity of the coordinate (any coordinate linearising the tilting action is unique up to affine maps), the constancy of the kinetic coefficient (homogeneity under displacement excludes position-dependent coefficients), and the exclusion of u-type coordinates (which have reachable boundaries)—are established as rigidity theorems in Paper II. The bound-state count $N = 3$ is invariant across the family $-\frac{1}{2}\, d^2/d\phi^2 - g\, H(\sigma(\phi))$ for $g \in [0.80, 1.50]$ (verified pointwise; certificate included).

Combining $V = -H$ as the potential with the kinetic term of the canonical normalisation ($g = 1$):

$$\left[-\frac{1}{2}\frac{d^2}{d\phi^2} - H(\sigma(\phi))\right]\Psi = E\,\Psi \tag{6}$$

This is nothing other than a Schrödinger equation: with only the binary nature (A1) and the uncertainty (A2, A3) of existence as input, the eigenvalue equation of quantum mechanics emerges as the continuum limit of the transfer matrix of the canonical action.

The bound-state count $N = 3$ is a consequence of this construction—the A3-selected coordinate ϕ together with a flat kinetic term. Writing the flat kinetic term in a different coordinate yields a different operator and a different spectrum, so the A3 selection of ϕ is essential.

The derivation chain of §2 is summarised as:

$$\text{A1–A3} \;\Rightarrow\; n^2{=}n,\ 0<\sigma<1 \;\Rightarrow\; Z = 1{+}e^{\phi} \xrightarrow{\text{Legendre}} V = -H$$
$$\xrightarrow{\text{transfer matrix}} \left[-\tfrac{1}{2}\partial_\phi^2 + V\right]\Psi = E\Psi$$

[2] The same normalisation is supported from the Landauer viewpoint. With ϕ as the e-affine coordinate, the transfer kernel takes the form $K(\phi', \phi; \Delta\tau) \propto \exp[-(\phi' - \phi)^2/(2\Delta\tau) - \Delta\tau V(\phi)]$; fixing the information metric per bit to unity gives kinetic coefficient 1. This is consistent with $g = 1$ (an independent corroboration; the proof of uniqueness is in Paper II).

3 Three Bound States

Why do fermions come in exactly three generations—why τ, μ, e, and why not a fourth? This chapter proves that the potential $V = -H$ has exactly three bound states (hereafter $N = 3$), and shows that this structurally corresponds to the three generations of fermions.

3.1 Why $N = 3$

The potential $V = -H$ is a well. It is deepest at the centre ($\phi = 0$) where $V = -\ln 2$, returning to zero at both ends $\phi \to \pm\infty$. In quantum mechanics, such a well confines discrete energy levels—by the same principle that the hydrogen atom confines electrons to discrete orbitals. A deep, wide well holds many levels; a shallow, narrow well holds few. The well of $V = -H$ is shallow—its depth is only $\ln 2 \approx 0.693$, and its shape is entirely determined by the Shannon entropy.

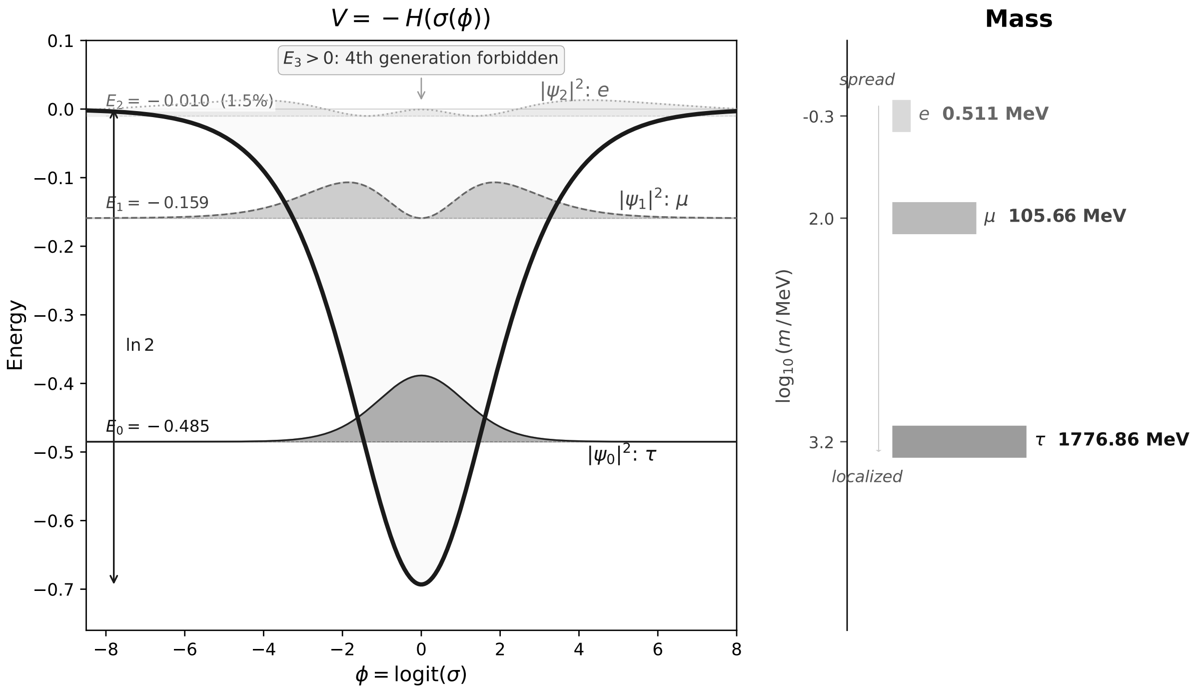

Figure 1: The potential $V = -H(\sigma(\phi))$ and the probability densities $|\psi_n|^2$ of its three bound states (left). More localised states correspond to heavier particles (right). $E_3 > 0$: a fourth generation is forbidden.

We denote the number of levels confined in this well by N. A semiclassical (WKB) approximation estimates how many half-wavelengths fit inside the well. Each integer half-wavelength corresponds to one bound state:

$$n_{\mathrm{WKB}} = \frac{1}{\pi} \int_{-\infty}^{\infty} \sqrt{2H(\sigma(\phi))}\, d\phi = 2.781\,. \tag{7}$$

The integer part is 2—two bound states are guaranteed. The fractional part 0.781 indicates that a third barely fits. A fourth would require $n_{\mathrm{WKB}} > 3$, but the well depth $\ln 2 \approx 0.693$ is insufficient. The WKB estimate suggests $N = 3$. The gap between the exact bound-state count and this semiclassical action count, $\varepsilon \equiv N - n_{\mathrm{WKB}} = 3 - 2.781 =$

0.219, is the spectral marginality of the barely bound third state; it controls the NLO corrections to the couplings and mixing angles derived in §6–§8. ε is defined by this specific action integral.[3] Different semiclassical conventions (e.g. with the Maslov correction) yield different action counts; throughout this paper ε always refers to the definition above.

$N = 3$ arises from a construction of three elements—the choice of coordinate, the normalisation of the kinetic term, and the type of entropy. The coordinate $\phi = \mathrm{logit}(\sigma)$ is uniquely determined by the exponential-family structure (§2.2), the kinetic-term normalisation is the canonical normalisation $g = 1$ of §2.4, and Shannon entropy is the unique entropy satisfying the Khinchin axioms, which include strong additivity (the chain rule) (Khinchin's theorem [16, 17]). $N = 3$ is rigorously proved by interval arithmetic under this construction (§3.2).

3.2 Rigorous Proof

Theorem 1 ($N = 3$)**.** *The Schrödinger equation of §2 has exactly three bound states, and their eigenvalues are rigorously enclosed by interval arithmetic:*

$$
\begin{aligned}
E_0 &= -0.485002253102964237445277765 8 \pm 10^{-24},\\
E_1 &= -0.15905222161885617799341 90 \pm 10^{-24},\\
E_2 &= -0.01042339306210945932761 6 \pm 10^{-22}
\end{aligned}
$$

No fourth negative eigenvalue exists; the essential spectrum begins at 0 (the small positive values seen on finite grids are discretisation artifacts of the box).

Proof (computer-assisted, interval arithmetic). By parity, decompose into the Neumann problem (even) and the Dirichlet problem (odd) on the half-line $[0, \infty)$. Under interval enclosures based on the monotonicity of the potential, propagate the Prüfer angle by intervals and combine with analytic tail estimates to obtain rigorous upper and lower bounds on the Prüfer angle at the trial energies. By Sturm oscillation theory, the number of negative eigenvalues is counted exactly by the rotation of the Prüfer angle, giving exactly 2 in the even sector and exactly 1 in the odd sector—$N = 3$ in total. The eigenvalue enclosures are certified by two-sided shooting with a verified high-order Taylor integrator over interval coefficients. An independent implementation (Arb ball arithmetic) re-certifies the bound-state count $N = 3$. The bundled verification code additionally confirms the existence of three negative eigenvalues by variational upper bounds from Rayleigh quotients of three sech-type (Pöschl–Teller-type [18]) trial functions. Three certificate scripts are included in the supplementary materials. $\square$

Correspondence with generations. These three bound states ψ_0, ψ_1, ψ_2 correspond to the three generations of fermions. The ordering of binding energies $|E_0| > |E_1| > |E_2|$ corresponds to the ordering of wavefunction localisation — the most deeply bound ψ_0 is the most localised at the centre of the well, while the marginally bound ψ_2 is the most spread out. As shown in §8, this localisation corresponds to mass, and ψ_0, ψ_1, ψ_2 correspond to the heaviest through the lightest generation (for charged leptons, τ, μ, e). The number of bound states (3) and their ordering structure correspond to the number of generations and the mass hierarchy. The absence of a fourth negative eigenvalue predicts that a fourth generation does not exist. The quantitative derivation of mass ratios and mixing angles is given in §8.

[3] The rigorous interval enclosure 0.219188 is given in Paper II.

4 Three Spatial Dimensions

Section 3 proved $N = 3$. The next question is: if the mathematical structure of these three bound states corresponds to physical structure, in how many spatial dimensions can they be realised? We now examine this.

Consider the same well shape $V = -H$ realised as a radial profile in d-dimensional space. Reducing the d-dimensional Schrödinger equation to a one-dimensional radial equation by standard partial-wave decomposition, an additional effective force $(d-1)(d-3)/(8r^2)$ appears:

$$V_{\text{eff}}(\phi) = -H(\sigma(\phi)) + \frac{(d-1)(d-3)}{8\phi^2}. \tag{8}$$

The numerator $(d-1)(d-3)$ vanishes for $d = 1$ and $d = 3$, leaving $V = -H$ unchanged. In other dimensions, this geometric term deforms the operator:

d	$(d-1)(d-3)/8$	Effect on the operator
1	0	No deformation
2	$-1/8$	An attractive term is added (tending to create extra states)
3	$\mathbf{0}$	**No deformation: the well shape is preserved**
4	$+3/8$	A repulsive term is added (tending to remove the marginal third state)
≥ 5	≥ 1	Strong repulsion

Full-line quantisation ($\phi \in \mathbb{R}$) is essentially self-adjoint at the endpoints and requires no boundary condition. Under A3 (non-existence is not a state—no additional data is admissible at the boundary; Paper II), this canonical full-line quantisation is the reference, and a spatial realisation is required not to add an extra inverse-square term to the canonical operator. This requirement is met only when the coefficient $(d-1)(d-3)/8$ vanishes, i.e. only for $d = 1, 3$. Note that a vanishing inverse-square term does not make the full-line and radial problems identical—a half-line radial model adds endpoint structure at the origin (a choice of self-adjoint extension) and is a different model, outside the comparison fixed by this correspondence (Paper II).

Two physical conditions exclude $d = 1$. First, the half-integer spin of fermions requires the spin-statistics theorem, which holds for $d \geq 3$ ($\pi_1(\mathrm{SO}(d)) = \mathbb{Z}_2$, giving the double cover $\mathrm{Spin}(d)$ with finite-dimensional spinor representations). Second, gravity propagating as a dynamical field requires $d \geq 3$ (for $d \leq 2$ the Weyl tensor vanishes identically, leaving no local degrees of freedom).

We summarise this as a theorem.

Theorem 2 ($d = 3$)**.** *Under the A3 selection of full-line quantisation, $d = 3$ is the unique spatial dimension simultaneously satisfying:* (i) *the well operator is preserved with no extra geometric term: $d = 1, 3$;* (ii) *spin-statistics connection holds: $d \geq 3$;* (iii) *gravity propagates (Weyl tensor non-trivial): $d \geq 3$. Intersection: $\{3\}$.*

Proof. (i) centrifugal term $(d-1)(d-3)/(8r^2) = 0$ iff $d = 1$ or 3; (ii) $\pi_1(\mathrm{SO}(d)) = \mathbb{Z}_2$ for $d \geq 3$; (iii) the Weyl tensor vanishes identically in spacetime dimension ≤ 3, i.e. spatial $d \leq 2$; propagating gravity requires $d \geq 3$. □

5 Gauge Structure from the Projective Plane

The generation number $N = 3$ (§3) and the spatial dimension $d = 3$ (§4) were derived independently—yet they take the same value $N = d = 3$. On this coincidence, we construct the three-dimensional space $\mathbb{F}_3^3$ over the field $\mathbb{F}_3$. This chapter shows that the geometric structure arising from it has the same mathematical construction as the Standard Model gauge structure.

5.1 Origin of the Gauge Structure

What gauge structure arises from $N = d = 3$? The key is the fundamental difference between generations and colour.

Quark colours (red, green, blue) are degenerate—every colour has the same energy, so there is no preferred basis. This equivalence generates the continuous rotation symmetry SU(3). Generations are different. The three bound states of $V = -H$ all have distinct energies ($E_0 \neq E_1 \neq E_2$, by the Sturm–Liouville theorem), so a preferred eigenbasis exists. The symmetry preserving this basis is discrete and, as a gauge symmetry, cannot support a continuous group.[4]

§3 derived $N = 3$ (bound states) and §4 derived $d = 3$ (spatial dimension). From this coincidence, the three-dimensional space $\mathbb{F}_3^3$ over the field $\mathbb{F}_3 = \{0, 1, 2\}$ is determined. The projective construction that follows requires a field (not merely a group); since $N = 3$ is prime, $\mathbb{F}_3 = \mathbb{Z}/3\mathbb{Z}$ is the unique field with N elements. Removing the origin and taking a two-dimensional cross-section gives $\mathbb{F}_3^2 = \{0, 1, 2\} \times \{0, 1, 2\}$—a 3×3 grid. These 9 points are the starting point for the construction.

Drawing lines on the 3×3 grid, lines with the same slope are parallel and do not intersect. In projective geometry, parallel lines meet at a single "point at infinity"—the same principle by which parallel railway tracks converge to a point on the horizon in perspective drawing. The lines on $\mathbb{F}_3^2$ have four slopes (horizontal, vertical, right-diagonal, left-diagonal), each generating one point at infinity. These four points form the "line at infinity" L—the horizon.

Adding the 4 points at infinity to the original 9 gives $9 + 4 = 13$ points. This is the projective plane $\mathrm{PG}(2, \mathbb{F}_3)$—the space obtained by defining "points" as one-dimensional subspaces and "lines" as two-dimensional subspaces through the origin of $\mathbb{F}_3^3$, with a symmetric structure of 13 points, 13 lines, 4 points per line, and 4 lines through each point (Figure 2).

Under the full symmetry $\mathrm{PGL}(3, \mathbb{F}_3)$ of $\mathrm{PG}(2, \mathbb{F}_3)$, all 13 points are equivalent and no partition arises. We now fix one of the 4 points on the line at infinity L as a reference point P_0—the pair (P_0, L) is called a flag. Fixing the flag is a choice of reference within the construction: $\mathrm{PGL}(3, \mathbb{F}_3)$ acts transitively on flags, so every choice yields an isomorphic partition. The fixing splits the 13 points into three groups: 9 points off L (the 3×3 grid), 3 points on L excluding P_0, and P_0 itself. Choosing P_0 alone gives only 1+12 = two groups; choosing a flag produces 9+3+1. This is the minimal structure distinguishing three forces:

$$\underbrace{N^2}_{\text{9 colour}} + \underbrace{N}_{\text{3 weak}} + \underbrace{1}_{\text{1 hyp}} = N^2 + N + 1. \tag{9}$$

[4] The $2C_F$ appearing in the mass readout of §8.1 is the internal-localisation Casimir of $N = 3$ and is compatible with this.

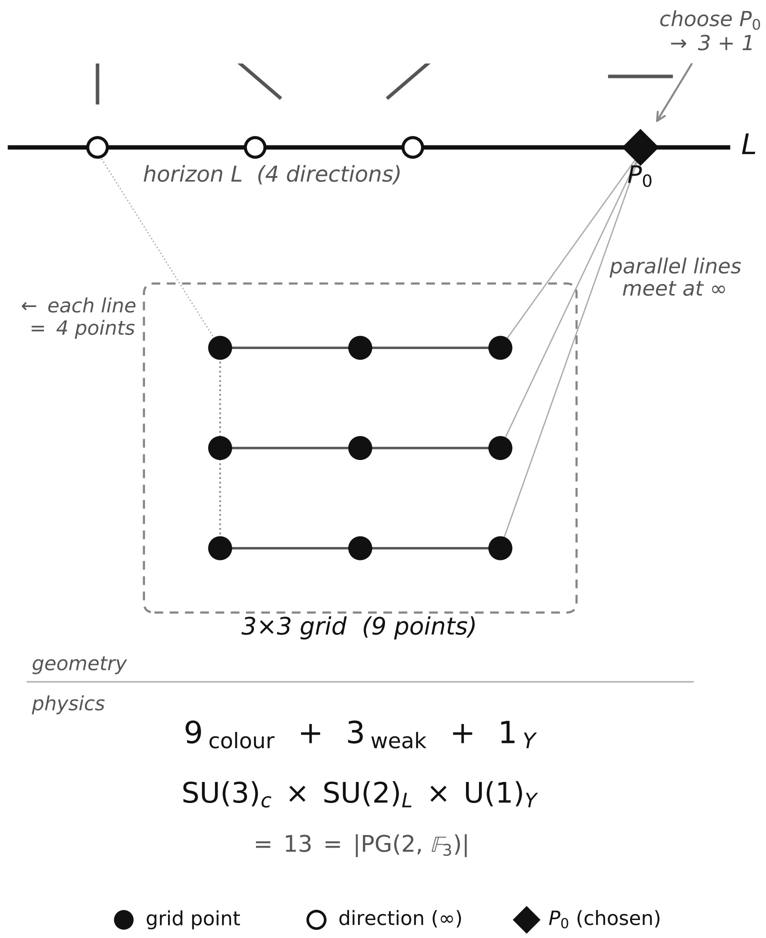

Figure 2: The projective plane $PG(2, \mathbb{F}_3)$. Parallel lines on the 3×3 grid (9 points, filled circles) meet at directions on the line at infinity L (open circles). Choosing P_0 (filled diamond) produces the partition $9+3+1 = 13$, corresponding to the sector structure of SU(3)×SU(2)×U(1).

$N = d$ is satisfied only by $N = 3$ ($N = 2, 4, 5$ all have $d = 3 \neq N$). The primality of $N = 3$ guarantees that the field with N elements is uniquely the prime field $\mathbb{F}_3 = \mathbb{Z}/3\mathbb{Z}$. All numbers in the partition 9+3+1 are determined by N alone: $N^2 = 9$, $N+1 = 4$ (excluding the reference point gives $N = 3$), reference point 1. Total: $N^2+N+1 = 13$. The 52 flags (point-line incidence pairs) provide the basis for the Laplacian (§6.3).

5.2 From 13 Points to the Standard Model Gauge Group

The partition 9+3+1 matches the sector structure of the Standard Model gauge group $SU(3)\times SU(2)\times U(1)_Y$.

9 points → SU(3). The 9 off-line points constitute the affine plane $AG(2, \mathbb{F}_3)$. These 9 points are labelled by $(a, b) \in \mathbb{F}_3^2$ and naturally index the finite Weyl operators $W_{\{a,b\}} = X^a Z^b$ in the three-dimensional internal representation. The identity $W_{\{0,0\}}$ is the central component; the remaining 8 non-trivial Weyl components form the traceless sector. By the Killing–Cartan classification, the unique rank-2, dimension-8 compact simple Lie completion is $\mathfrak{su}(3)$. The 9th, central component corresponds to the global baryon number

$\mathrm{U}(1)_B$ (an ungauged global charge; the exact anomaly-free combination is $B{-}L$, §8.4).
3 points → SU(2). The 3 points on L excluding P_0 give three non-trivial weak directions relative to the reference point. The minimal non-abelian compact Lie completion of these three directions is the 3-generator $\mathfrak{su}(2)$, whose doublet representation gives the weak SU(2) action.
1 point → $\mathbf{U(1)}_Y$. The reference point P_0 corresponds to the abelian central module phase, and physically to $\mathrm{U}(1)_Y$.
Direct product guarantee. Translations of the affine plane fix the line at infinity L—shifting the grid does not change the "directions." The colour sector (9 points) transformations do not mix with the electroweak sector (3+1 points). This is a geometric property following from the incidence structure of $\mathrm{PG}(2, \mathbb{F}_3)$: it yields a direct-product structure of three mutually non-mixing sectors—the same shape as the Standard-Model gauge group. The global form of the group (in the Standard Model, including the $\mathbb{Z}_6$ quotient [38]) is not fixed by the geometry alone; it is determined by the faithful action on the particle carrier (Paper II).

The complete derivation chain is:

$$\begin{aligned} n^2{=}n \xrightarrow{V=-H} N{=}3 \xrightarrow{d=3} \mathbb{F}_3^3 \xrightarrow{\text{count directions}} 13 \text{ pts} \\ \xrightarrow{\text{flag}} 9{+}3{+}1 \xrightarrow{\text{Killing–Cartan}} \mathfrak{su}(3){\oplus}\mathfrak{su}(2){\oplus}\mathfrak{u}(1). \end{aligned} \tag{10}$$

This mathematical result agrees with the experimentally established gauge structure $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$.

The 4 on-line points correspond to the four real degrees of freedom of the SU(2) doublet — the three Goldstone bosons required for $\mathrm{SU}(2){\times}\mathrm{U}(1) \to \mathrm{U}(1)_{\mathrm{em}}$ breaking and the physical Higgs. $V = -H$ takes its minimum $-\ln 2$ at $\phi = 0$ ($\sigma = 1/2$) and returns to zero as $\phi \to \pm\infty$. This shape defines a stable well ($V''(0) = 1/4 > 0$). The vacuum sits at the interior point $\sigma = 1/2$ rather than at the vanishing-field points $\sigma \in \{0, 1\}$—the role played in the Standard Model by the negative quadratic term ($\mu^2 > 0$), moving the vacuum from the symmetric point to a finite field value, is here played by the shape of the well.

Configuration decomposition of the existence variable n. The gauge structure creates diverse configurations within the internal space of the existence variable n (§2). The variable n is unique — by A3 (indiscernible ⇒ identical), bare existence with no distinguishing properties collapses to one. But the $\mathrm{PG}(2, \mathbb{F}_3)$ gauge structure defines multiple internal configurations: each configuration i is characterised by its generation (bound state of $V = -H$) and gauge quantum numbers (position in PG), and carries an occupation number $n_i \in \{0, 1\}$. n is one, n_i are many — a unique existence manifests as diverse fermions. The configuration decomposition is:

$$\sum_i \hat{n}_i = \hat{n}, \qquad \hat{n}_i^2 = \hat{n}_i, \qquad \hat{n}_i\, \hat{n}_j = \delta_{ij}\, \hat{n}_i. \tag{11}$$

The first equation states that existence decomposes into configurations; the second that bivalence ($n^2{=}n$) is inherited by each configuration; the third that the unique unit of existence occupies exactly one configuration—the orthogonal resolution of the one-particle state space (the $\hat{n}_i$ are mutually orthogonal projections summing to $\hat{n}$). This mutual exclusivity is not an independent assumption but a consequence of the bivalence of existence (A1) and its uniqueness (A3). This is one-particle-level structure: the Pauli exclusion principle itself—no double occupancy of a single mode, together with exchange antisymmetry—is carried by the many-body realisation (the CAR construction, Paper II), where distinct configurations can be simultaneously occupied. The concrete configuration labels are fixed by the fermion representations in §5.3. $n_i = 0$ does not mean non-existence; it means existence occupies a different configuration.

5.3 Origin of Fermion Diversity

§5.2 established that the 13 directions of $\mathrm{PG}(2, \mathbb{F}_3)$ establish a structure corresponding to the gauge group SU(3)×SU(2)×U(1). But the 13 directions do not correspond to fermions themselves. The 13 directions define the gauge symmetry — the structure of forces — and the fermion structure emerges as **representations** of this gauge group. Here we show why the single existence variable n gives rise to a diverse matter-particle structure.

Before the gauge structure emerges, n has no quantum numbers — no generation label, no colour, no isospin. Without distinguishing properties, multiple instances of n are identical (Leibniz's principle of the identity of indiscernibles), and n remains unique. The gauge structure of $\mathrm{PG}(2, \mathbb{F}_3)$ changes this: the representations of the gauge group create diverse "slots" — configurations — in the internal space, with each configuration carrying an occupation number $n_i \in \{0, 1\}$.

Each configuration i is characterised by two labels: generation k (which bound state of $V = -H$, $k = 1, 2, 3$) and gauge representation α (which sector of $\mathrm{PG}(2, \mathbb{F}_3)$). That is, $n_i = n_{k,\alpha}$. The representations of SU(3)×SU(2)×U(1) give 16 states per generation. The chiral asymmetry (SU(2) acting only on left-handed fields) is part of this particle-species specification and is not derived in this paper. In Paper II, under the stated enumeration conditions, mixed configurations are excluded and the left/right assignment is unique up to a global orientation convention. The hypercharge values in the table below are the unique anomaly-free assignments compatible with the 9+3+1 sector structure, Yukawa closure, the inclusion of ν_R, and the specification of a single local abelian gauge factor ($B{-}L$ remains an ungauged global charge, §5.2); the operator-level derivation is likewise given in the same paper:

Representation	Content	States
$(\mathbf{3}, \mathbf{2}, 1/6)$	Left-handed quarks Q_L	6
$(\mathbf{3}, \mathbf{1}, 2/3)$	Right-handed up-type u_R	3
$(\mathbf{3}, \mathbf{1}, -1/3)$	Right-handed down-type d_R	3
$(\mathbf{1}, \mathbf{2}, -1/2)$	Left-handed leptons L_L	2
$(\mathbf{1}, \mathbf{1}, -1)$	Right-handed charged lepton e_R	1
$(\mathbf{1}, \mathbf{1}, 0)$	Right-handed neutrino ν_R	1

Total: 16×3 generations $= 48$ configurations. The inclusion of ν_R is required for Yukawa closure and for the $B{-}L$-protected Dirac-neutrino structure (see §8.4 of this paper). This is the concrete realisation of the configuration decomposition above: the index i runs over the pair (k, α), where k is the generation label and α is the gauge-representation label. The inherited bivalence $n_i^2 = n_i$ is what, in the many-body realisation, restricts the occupation of each mode to $\{0, 1\}$—this is the occupation-number representation of the Pauli exclusion principle [11]. The mutual exclusivity of distinct configurations is the one-particle-level structure of §5.2: which configuration the unique unit of existence occupies. The generators of SU(3)×SU(2)×U(1) correspond to 8+3+1 = 12 gauge bosons

(8 gluons, 3 weak bosons W^+W^-Z, 1 photon). Together with the Higgs boson from §5.2, the 48 fermions, 12 gauge bosons, and 1 Higgs constitute the complete particle content of the Standard Model.

Thus, diverse matter (n_i) arises from unique existence (n). The key is that the gauge structure of $\mathrm{PG}(2, \mathbb{F}_3)$ creates "distinguishable slots" — without gauge symmetry, existence remains featureless and singular; with gauge symmetry, 48 species of fermions emerge. The diversity of matter is a consequence of gauge structure.

The internal consistency of this construction is independently confirmed by the eigenfunction participation ratio $D_{PR} = (\int \rho \, d\phi)^2 / \int \rho^2 \, d\phi$, with $\rho = \sum_n \psi_n^2$ the total density of the three bound states, $= 13.06 \approx |\mathrm{PG}|$ (reproduced by the verification code) and by the spectral decomposition of the 52×52 flag Laplacian.[5] No step in the derivation contains a free parameter. The validity of this construction is supported by the fact that all independent physical quantities derived in §6–10 agree with experiment to 0.0006%–2.6% accuracy.

6 Gauge Coupling Constants

§5.2 derived the structure corresponding to the gauge group (which forces exist). This chapter derives their strengths (coupling constants).

6.1 Weinberg Angle

The partition 9+3+1 of §5.2 gives values corresponding to the coupling constants. The fraction of the 13 directions occupied by the 3 weak-sector directions is the Weinberg angle [21] (weak mixing angle) $\sin^2\theta_W$. $\mathrm{PGL}(3, \mathbb{F}_3)$ acts transitively on $\mathrm{PG}(2, 3)$, so all directions carry equal weight (Haar measure). The Weinberg angle is therefore a simple fraction:

$$\sin^2\theta_W = \frac{N}{N^2+N+1} = \frac{3}{13} = 0.2308 \qquad \text{(expt: } 0.23122 \ \overline{\mathrm{MS}} \text{ at } M_Z, \ 0.2\%\text{).} \tag{12}$$

The tree-level prediction $3/13 = 0.23077$ deviates from the $\overline{\mathrm{MS}}$-scheme experimental value 0.23122 ± 0.00003 [22] by 0.2%, consistent with one-loop radiative corrections. This gives the tree-level W/Z mass ratio: $m_W/m_Z = \cos\theta_W = \sqrt{10/13} = 0.8771$ (expt: 0.8813, 0.5%; deviation consistent with radiative corrections).

6.2 Strong Coupling Constant

The strong coupling constant also follows from the colour-sector fraction. Of the 9 colour-sector directions, the independent directions number $\mathrm{rank}(\mathrm{SU}(3)) = 2$ (Cartan generators), and by equipartition under the Haar measure, the coupling constant per direction is $\sin^2\theta_W$ divided by 2: $\alpha_s^{(0)} = \sin^2\theta_W/(N-1) = 3/26 = 0.1154$. The spectral marginality $\varepsilon = 0.219$ from §3 provides the NLO correction. The correction is distributed over $N^2+1 = 10$ non-weak PG directions (colour 9 + hypercharge 1):

$$\alpha_s = \frac{3}{26}\left(1 + \frac{\varepsilon}{N^2+1}\right) = 0.1179 \qquad \text{(expt: } 0.1179 \ \overline{\mathrm{MS}} \text{ at } M_Z, \ 0.01\%\text{).} \tag{13}$$

[5] The finite-kernel theorem and its spectral gauge-decomposition corollary in Paper II.

The value $k = N^2{+}1 = 10$ follows from the number of non-weak directions in $\mathrm{PG}(2, \mathbb{F}_3)$. Each k-value is determined as the geometric range of the Borel subgroup of $\mathrm{PGL}(3, \mathbb{F}_3)$ acting on the corresponding sector—$k = N^2 = 9$ for generation mixing (affine subgroup), $k = N^2{+}1 = 10$ for the non-weak sector (affine + hypercharge), $k = |\mathrm{PG}| = 13$ for the Laplacian (full projective space).

6.3 Fine-Structure Constant

The Weinberg angle and strong coupling constant were determined by simple point ratios on $\mathrm{PG}(2, 3)$. The fine-structure constant requires more—the full spectral structure of the internal space.

The inverse electromagnetic coupling $1/\alpha$ measures the weakness of the electromagnetic interaction. The observed charge is the charge screened by the matter fields—the stronger the screening, the weaker the coupling and the larger $1/\alpha$. In this framework $1/\alpha$ is given (up to the self-consistent correction $-\alpha$) by the sum of the bare photon propagation cost λ_1 and the matter screening quantity Z, and the Thomson-point value is dominated by the screening term. In 4D QED this structure is formalised as the Dyson equation. In 0D the same structure holds but becomes algebraically exact—the propagator is a constant, so the vertex correction and self-energy are determined by the same resolvent, and the 4D perturbative cancellation $Z_1 = Z_2$ holds as an exact algebraic identity (proof in Paper II). The self-consistency equation is:

$$\frac{1}{\alpha} + \alpha = \lambda_1 + Z. \tag{14}$$

The right-hand side has two terms: λ_1 is the bare photon inverse propagator (propagation cost) and Z corresponds to the scalar-channel spectral quantity (screening). This correspondence is a spectral one—Z is the resolvent trace whose energy-denominator structure matches the matter vacuum polarisation. The form of the equation (the left-hand side $1/\alpha + \alpha$ with unit coefficient) belongs to the specified correspondence of the incidence current with the Thomson-point electromagnetic current. The full justification of this correspondence as a self-consistent prediction, and the proof that the physical root is unique under this reading, are both given in the same paper. The dominant contribution to Z is $1/|E_2| = 95.94$; since E_2 is rigorously certified by the interval arithmetic of §3.2, this sensitivity introduces no uncertainty.

The minimum cost λ_1 for gauge-field propagation through internal space is the smallest non-zero eigenvalue of the Laplacian on internal space. Since gauge interactions involve both positions (points) and directions (lines), the natural object for diffusion is a flag (point-line incidence pair). The spectrum of the graph Laplacian L on the flag graph of $\mathrm{PG}(2, \mathbb{F}_3)$, a 52×52 matrix, is:

Eigenvalue	Value at $N = 3$	Multiplicity
0	0	1
$(N{+}1) - \sqrt{N}$	2.268	12
$(N{+}1) + \sqrt{N}$	5.732	12
$2(N{+}1)$	8	27

The smallest non-zero eigenvalue $\lambda_1 = 4 - \sqrt{3} = 2.268$ is the quantity that corresponds to the bare photon inverse propagator in the $1/\alpha$ correspondence (§6.3). $\lambda_1 > 0$ is consistent with the confinement picture—the positive spectral gap corresponds to colour-charged states not propagating freely in the asymptotic regime.

The screening term Z corresponds to the scalar-channel spectral quantity from three generations of matter fields, determined by the resolvent of the internal kinetic operator. The internal kinetic operator is a tensor product of the generation sector (binding energies $|E_n|$) and the gauge sector (Laplacian eigenvalues λ_k):

$$K_{\text{int}} = K_{\text{gen}} \otimes \mathbf{1} + \mathbf{1} \otimes L$$

This form—the tensor sum, with both factors measured in the same unit at relative scale 1—is established in Paper II as a uniqueness theorem under the single-clock unit clause (product couplings, twisted products, and rescalings of the relative unit are excluded). Decomposing the trace of the resolvent by multiplicity:

$$Z = \text{Tr}[1/K_{\text{int}}] = \zeta_V(1) + 12\, S_1 + 12\, S_2 + 27\, S_3 = 134.78$$

where $\zeta_V(1) = \sum_n |E_n|^{-1}$ and $S_i = \sum_n (|E_n| + \lambda_i)^{-1}$. The intermediate values are: $\zeta_V(1) = 104.29$ (dominated by $E_2^{-1} = 95.94$), $12S_1 = 14.57$, $12S_2 = 6.06$, $27S_3 = 9.86$. The 27-dimensional sector (multiplicity $27 = 3^3$) contributes $9.86 \approx N^2$, providing the spectral structure corresponding to the QCD vacuum-polarisation correction (all intermediate values are reproducible with the accompanying verification code).

Substituting $\lambda_1 = 2.268$ and $Z = 134.78$ into the self-consistency equation:[6]

$$\frac{1}{\alpha} = \frac{(Z+\lambda_1) + \sqrt{(Z+\lambda_1)^2 - 4}}{2} = 137.0368 \quad (\text{expt: } 137.036,\ 0.0006\%). \tag{15}$$

7 Higgs Sector and Electroweak Symmetry Breaking

§6 derived the gauge coupling constants. The Standard Model has one more free parameter in the scalar sector — the Higgs self-coupling λ. This chapter derives the mathematical structure corresponding to it and shows the pinning of the internal vacuum and the boundedness of the internal potential (no instability channel).

7.1 Symmetry Breaking and Higgs Mass

Electroweak symmetry breaking is the statement that, while the dynamical laws remain symmetric, the vacuum (ground state) departs from the symmetric point. Since the W and Z are massive while the photon is massless, the vacuum cannot sit at the field-vanishing point (the symmetric point)—breaking is required by experiment. The Standard Model realises this through the assumption $\mu^2 > 0$ in the Higgs potential $V = -\mu^2|\Phi|^2 + \lambda|\Phi|^4$: it destabilises the origin, rolling the field to a finite value. But "why $\mu^2 > 0$" is not explained within the Standard Model.

The $V = -H(\sigma(\phi))$ of this paper answers this question through structure. The binary Shannon entropy H vanishes at both endpoints $\sigma \in \{0, 1\}$ (the trivial configurations corresponding to the field-vanishing point) and is maximal at the interior point $\sigma = 1/2$, so $V = -H$ is a stable well that is 0 at the endpoints and reaches its minimum $-\ln 2$

[6] The recommended experimental value is $1/\alpha = 137.035999177(21)$ (CODATA 2022). The value 137.0368 in the text is the leading term; the high-precision evaluation including the correction series is treated in Paper II.

at the interior point ($V''(0) = 1/4 > 0$). Moreover A2 ($\sigma \neq 1$) and A3 ($\sigma \neq 0$) forbid the endpoints themselves. The ground state therefore necessarily sits at the interior point—the condition that the vacuum cannot sit at the symmetric point, i.e. the condition for breaking, follows from the shape of the entropy and the axioms rather than from destabilising the origin. The position of the vacuum is a consequence of the shape of V and requires no additional assumption. The readout of the Higgs mass uses the positive curvature $V''(0) = 1/4$ at this minimum.

§5.2 showed that the line structure of $\mathrm{PG}(2, \mathbb{F}_3)$ corresponds to the Higgs SU(2) doublet. The curvature at the centre gives the Higgs mass: $V''(0) = g_F(0) = \sigma(0)(1-\sigma(0)) = 1/4$ (the Fisher metric at $\sigma = 1/2$).

$$m_H/v = \sqrt{V''(0)} = 1/2 \quad \text{(LO)}. \tag{16}$$

The NLO correction enters from all $|\mathrm{PG}| = 13$ modes of the Laplacian matrix L:

$$m_H/v = \frac{1}{2}\sqrt{1 + 2\varepsilon/|\mathrm{PG}|} = 0.5084 \quad \text{(experiment: } 0.5082,\ 0.03\%\text{)}. \tag{17}$$

This corresponds to $\lambda = (m_H/v)^2/2 = 0.1292$ (expt: 0.1291, 0.06%).

7.2 Vacuum Stability and the Hierarchy Problem

The Standard Model Higgs potential $V_{\mathrm{SM}} = -\mu^2 h^2 + \lambda h^4$ diverges as $h \to \infty$, and the UV correction $\delta\mu^2 \propto \Lambda^2$ requires $\sim 10^{34}$ fine-tuning between the electroweak and Planck scales (hierarchy problem). Furthermore, the Higgs quartic coupling $\lambda(\mu)$ runs negative at $\mu \sim 10^{10}$ GeV, making the electroweak vacuum cosmologically metastable.

In the internal sector of $V = -H$, the instabilities corresponding to these problems do not arise. $V = -H$ is bounded ($|V| \leq \ln 2$), with no high-energy divergence. The well depth $|V_{\min}| = \ln 2$ and curvature $V''(0) = 1/4$ are both $O(1)$ algebraic constants; their ratio gives the electroweak scale $v^2/\Lambda^2 = |V_{\min}|/V''(0) = 4 \ln 2 \approx 2.77$, an order-unity number requiring no fine-tuning. The internal potential $V = -H$ is bounded: $V \in [-\ln 2, 0]$. Interpreting this boundedness as the reason the running coupling $\lambda(\mu)$ in 4D does not turn negative requires the reconstruction matching (likewise the full 4D vacuum-stability interpretation and the correspondence to thermal properties such as symmetry restoration at high temperature, Paper II). The shape of V is fixed by the axioms, and scale dependence does not alter the algebraic structure of V. Furthermore, $V = -H$ is uniquely determined as the canonical free energy of $n \in \{0, 1\}$ (§2). Within the minimal PFE operator algebra, no second fundamental scalar potential is generated. The minimal reconstruction thus predicts a single Higgs doublet.

Radiative corrections. The predictions above are algebraic values, and their comparison with M_Z-scale experiments requires the 4D RG evolution and loop corrections of the running quantities (§9.3). The next correction to the weak angle is fixed in closed form by the framework's own readout: $\Delta_{\mathrm{rad}} = \alpha \cdot 135/2197 = 0.00045$ (derived in the same paper under the specified readout). This gives $\sin^2\theta_W = 3/13 + \Delta_{\mathrm{rad}} = 0.23077 + 0.00045 = 0.23122$, in agreement with the $\overline{\mathrm{MS}}$ value 0.23122 ± 0.00003 [22] within errors. Similarly, the leading value $m_W/m_Z = \sqrt{10/13} = 0.8771$ is 0.5% below experiment; this residual has approximately the size and sign of the standard SM top-quark $\Delta\rho$ correction, but the $\Delta\rho$-corrected comparison is a conditional computation requiring renormalisation-scheme matching.

Summary. $\sin^2\theta_W(M_Z) = 0.23122$ (0.2%), $\alpha_s(M_Z) = 0.1179$ (0.01%), $1/\alpha_{\mathrm{em}} = 137.0368$ (0.0006%), $m_H/v = 0.5084$ (0.03%). All four coupling constants are derived from $V = -H$ and $\mathrm{PG}(2,3)$ and agree with experiment. The vacuum is pinned at an interior point, the internal potential is bounded (no instability channel), and there is only one Higgs. The correspondence to 4D vacuum stability and the hierarchy problem is a matter of reconstruction matching.

8 Fermion Masses and Mixing

§5–7 established the structures corresponding to gauge structure, coupling constants, and the Higgs mass. Based on the correspondence between the three bound states established in §3 and the three generations of charged leptons, this chapter derives formulas for the mass ratios and mixing angles from the localisation of the bound states. The derived values agree to high precision with the mass ratios of charged leptons, quarks, and neutrinos, and with the CKM/PMNS mixing angles.

8.1 Mass Formula

The three bound states of $V = -H$ differ in localisation. The deepest ψ_0 is localised at the well bottom and couples most strongly to the gauge field, corresponding to the heaviest τ:

Bound state	Energy	Localisation	Particle	Mass (MeV)
ψ_0	$E_0 = -0.485$	High (well bottom)	τ	1777
ψ_1	$E_1 = -0.159$	Medium	μ	106
ψ_2	$E_2 = -0.010$	Low (well edge)	e	0.511

The mass ratios are determined by the localisation of the wave functions, not as independent parameters. The key quantity is $R = \ln(m_\tau/m_e)/\ln(m_\mu/m_e)$, which compresses the three-body mass hierarchy into a single number.

$$V = -H \xrightarrow{\S 3} \psi_n \xrightarrow{\mathrm{IPR}} \text{localisation} \xrightarrow{2C_F} m_n \rightarrow R = 1.5293 \tag{18}$$

Each eigenstate has a binding energy $|E_n|$ and a localisation. Localisation is measured by the inverse participation ratio $\mathrm{IPR}_n = \int |\psi_n|^4\, d\phi$. The coupling strength to the gauge field is proportional to the wave-function amplitude at the interaction point. A more localised state in internal space has a larger amplitude at that point and acquires a larger mass via the gauge field. Mass is proportional to a power of localisation: $m_n \propto \mathrm{IPR}_n^b$. In this section's readout, the exponent b is taken to be the internal-localisation Casimir of the continuous SU(3) lift associated with the $N = 3$ sector: $b = 2C_F = (N^2-1)/N = 8/3$.[7]

A small correction arises from the anharmonicity of $V = -H$. Each mode samples the well differently; the virial ratio $|\langle V\rangle_n|/T_n$ (ratio of potential to kinetic energy) measures this. The correction exponent is determined by the spectral ζ-function through the marginality parameter: the semiclassical action count $n_{\mathrm{WKB}} = 2.781$ falls short of the exact bound-state count $N = 3$ by $\varepsilon = 0.219$, giving a per-mode marginality $\delta = \varepsilon/N = 0.073$.

[7] This is not the physical colour charge of leptons but the $N = 3$ internal-localisation Casimir; for quarks the same $N = 3$ structure coincides with colour. See Paper II [20].

The complete mass formula is:

$$m_n \propto |E_n| \times \mathrm{IPR}_n^{2C_F} \times \left(\frac{|\langle V\rangle_n|}{T_n}\right)^{\varepsilon/N}. \tag{19}$$

The three factors have different origins:

Factor	Physical meaning	Determined by
$\lvert E_n\rvert$	Energy scale	Binding energy
IPR^{2C_F}	Localisation → coupling strength	Group theory (Casimir)
$(\lvert\langle V\rangle\rvert/T)^{\varepsilon/N}$	Anharmonic correction	Spectral marginality

All exponents are determined by $V = -H$. The result:[8]

$$R = \frac{\ln(m_\tau/m_e)}{\ln(m_\mu/m_e)} = 1.5293 \qquad \text{(experiment: 1.5294, 0.006\%)}. \tag{20}$$

Group theory fixes $b = 2C_F = 8/3$ first. Substituting this fixed value, both the Koide [23] constraint $Q \approx 3/2$ (our convention is $Q = (\sum\sqrt{m})^2/\sum m$, the inverse of the standard convention $\sum m/(\sum\sqrt{m})^2 = 2/3$) and the spectral condition $c = \varepsilon/N$ are independently satisfied. Koide is therefore not an input that determines b, but an output check of $b = 8/3$:

b	$c\,(Q{=}3/2)$	Deviation
2.0	0.437	500%
2.5	0.163	120%
$\mathbf{8/3 = 2C_F}$	**0.0730**	**0.07%**
2.78	0.012	84%
3.0	−0.107	246%

The individual mass ratios:

	Prediction	Experiment	Accuracy
m_τ/m_e	3481	3477	0.1%
m_μ/m_e	207.0	206.8	0.1%
R	1.5293	1.5294	0.006%
Q (Koide)	1.499985	1.50001(2)	2σ (Belle II)

The three lepton masses are not independent parameters but three modes of a single well, with the exponent $2C_F = 8/3$ fixed by this section's readout. In 0D there are no loop integrals, so the anomalous dimension is the internal-localisation Casimir of the continuous SU(3) lift. The Peter–Weyl theorem for finite groups guarantees that the heat kernel on the group takes the form $K(t) = \sum_\rho \dim(\rho)\,\chi_\rho(g)\,e^{-C_\rho t}$, admitting only exponential decay with no power-law terms. Consistency with the finite-group heat kernel supports this readout.[9] The 0.006% agreement of $R = 1.5293$ quantitatively supports the correspondence between bound states and lepton generations. The stability of the Koide relation under radiative corrections is discussed by Sumino [43].

[8] All spectral quantities are computed with $|\phi|_{\max} = 60$, $\Delta\phi = 2.34 \times 10^{-4}$ ($N_{\mathrm{grid}} = 512001$), verified stable under doubling of grid resolution.

[9] The uniqueness of this readout is treated in Paper II: the functional form of the mass response is proved as a uniqueness theorem, and the exponent assignment is treated as an explicitly specified readout.

8.2 Quark Masses

The mass formula of §8.1 reproduced $R = 1.5293$ for leptons. Can the same formula be applied to quarks? The formula itself does not change—only the well depth changes.

Leptons are colour singlets and feel $V = -H$ as a single copy. Quarks carry three colours. The colour-triplet well is taken to be $V = -3H$, the sum of the three isomorphic colour contributions on the common existence coordinate ϕ—a readout specification treated in Paper II. The deeper well supports $N = 5$ bound states ($n_{\mathrm{WKB}} = 4.82$):

Sector	Potential	c	Bound states	Modes used
Lepton	$-H$	1	3	$(0, 1, 2) = \tau, \mu, e$
Up-type	$-3H$	3	5	$(1, 2, 3) = t, c, u$
Down-type	$-3H$ + correction	3	4	$(0, 1, 2) = b, s, d$

Up-type. Mode 0 is deeply confined and belongs to the $I_3 = -1/2$ sector. Up-type quarks ($I_3 = +1/2$) use modes $(1, 2, 3)$, giving $R_{\mathrm{up}} = 1.777$ (experiment: 1.770, 0.4%).

Down-type. Up-type and down-type belong to the same SU(2) doublet and feel the same $V = -3H$, but differ in the sign of weak isospin ($I_3 = +1/2$ vs $-1/2$). This sign supplies a correction to the mass formula.

The total gauge charge of the SU(2) doublet (u, d), $2C_F + 1/N_c = N_c$, is an exact algebraic identity. The physical mechanism originates from $\mathrm{PG}(2, \mathbb{F}_3)$ geometry: among the $N{+}1 = 4$ points on line L, the reference point P_0 defines $I_3 = +1/2$, and the remaining 3 points generate SU(2). The isospin flip $P_0 \leftrightarrow L\backslash P_0$ reverses the SU(2) action, coupling the total charge N_c to $g_F(\phi) = \sigma(1{-}\sigma)$ (the Bernoulli Fisher metric $g_F = \mathrm{Var}(n)$ of §2.2; its uniqueness is Čencov's theorem [13]) with a negative sign. The effective mass is:

$$m_{\mathrm{eff}}(\phi) = 1 - N_c\, \sigma(1{-}\sigma) \tag{21}$$

This is the static Fisher-limit form of the down-sector readout. The operator actually solved is the symmetric position-dependent-mass form $-\frac{1}{2}\,\partial_\phi\,(1/m_{\mathrm{eff}})\,\partial_\phi - 3H$ (identical to the discretisation in the accompanying code). The correction coefficient $g = -N_c = -3$ is fixed by the algebraic structure of the SU(2) doublet. The corrected operator has 4 negative eigenvalues (-1.5709, -0.5455, -0.1952, -0.00733; stable for box size $L = 40/80/120$), of which 3 modes $(0, 1, 2)$ correspond to the physical window b, s, d—the same type of window rule by which the up-type sector selects $(1, 2, 3)$ from its 5 modes. With modes $(0, 1, 2)$: $R_{\mathrm{down}} = 2.273$ (experiment: 2.276, 0.1%).

The mass ratios of all three charged-fermion sectors are reproduced:

Sector	c	g	R (pred.)	R (expt.)	Accuracy
Lepton	1	0	1.529	1.529	0.006%
Up	$N_c = 3$	0	1.777	1.770	0.4%
Down	$N_c = 3$	$-N_c$	2.273	2.276	0.1%

Here R captures the shape of the log-hierarchy—it is invariant under a uniform stretching of the mass ratios (raising every m_i/m_j to a common power)—not the individual masses. The accuracy column gives the deviation of the prediction from the experimental central value. The experimental R inherits the light-quark mass uncertainties (PDG 2026: $m_u = 2.16 \pm 0.07$ MeV, etc.), which propagate to a width of about ± 0.15–0.25%, and the comparison further depends on the scheme and scale conventions for each mass (light quarks $\overline{\text{MS}}$ at 2 GeV, c and b at $m(m)$, t from direct measurement). Which of the modes correspond to which quarks is a physical specification (which 3 of the 5 modes), not a derivation.[10]

8.3 CKM Mixing

§8.1–8.2 established the mass ratios. With masses determined, we now derive the inter-generation mixing [24]. Why is V_{ub} small, and why does the Cabibbo angle [19] take its particular value? The Standard Model does not explain these. In $V = -H$, the potential symmetry corresponds to the former and the marginality of the third generation to the latter.

The potential $V = -H$ is symmetric: $V(\phi) = V(-\phi)$. This fixes the parity of each eigenfunction: ψ_0 even, ψ_1 odd, ψ_2 even. The direct overlap between distinct modes vanishes by Sturm–Liouville orthogonality: $\langle\psi_0|\psi_2\rangle = 0$. Since ψ_0 and ψ_2 are both even, $\langle\psi_0|M|\psi_2\rangle = 0$ for a V-parity-odd mixing operator M—the leading $0 \leftrightarrow 2$ transition is forbidden:

$$V_{ub}\big|_{\text{tree}} = 0 \tag{22}$$

The observed value $|V_{ub}| \approx 0.004$ arises at higher orders of the Wolfenstein hierarchy [25]. V-parity makes the tree/direct term exactly zero and suppresses distant-generation transitions to at least $O(\varepsilon^3)$; in the Paper II readout, the first non-zero term is $2\varepsilon^4$ (see below).

The Cabibbo angle follows from spectral marginality. The third bound state ψ_2 is marginally bound—its binding energy $|E_2| = 0.010$ is only 1.5% of the well depth. The parameter measuring this marginality is the gap between the exact bound-state count and the semiclassical action count: $\varepsilon = N - n_{\text{WKB}} = 3 - 2.781 = 0.219$.

Mixing angles reflect the mismatch between mass and weak eigenstates, determined by overlap integrals of different-mode wave functions. The marginality of the third bound state ($\varepsilon > 0$) extends the wave-function tails, generating overlap with adjacent modes. $\varepsilon \to 0$ removes the marginal mixing source, reducing to two generations and zero flavour transitions. To second order:

$$\sin\theta_C = \varepsilon\Big(1 + \frac{\varepsilon}{N^2}\Big) = 0.2245 \qquad \text{(experiment: } 0.2243,\ 0.10\%\text{)}. \tag{23}$$

This value is the unscreened internal value. The observed readout composed with colour screening is $\sin\theta_C \cdot (1 - x) = 0.22435$ (same paper).

The Cabibbo angle is a direct expression of the fact that the third generation barely exists: the generation-sector spectral quantity $\varepsilon = 0.219$, defined in §3 from $V = -H$ alone and entering all mixing readouts in common. If the third bound state were slightly more deeply bound ($\varepsilon \to 0$), inter-generation mixing would vanish and all flavour transitions would be forbidden.

The Wolfenstein hierarchy follows from the Cabibbo angle [25]: $|V_{us}| \sim \varepsilon$, $|V_{cb}| \sim \varepsilon^2$, and $|V_{ub}|$ is suppressed to at least $O(\varepsilon^3)$ with first non-zero term $2\varepsilon^4$ (experiment: 0.225, 0.041, 0.004). NLO corrections take the form $1 \pm \ell\varepsilon/k$. The denominator k is determined by the PG$(2, \mathbb{F}_3)$ geometry— $k = N^2 = 9$ for generation mixing ($\sin\theta_C$), $k = N^2{+}1 = 10$ for the non-weak sector (α_s), and $k = N = 3$ for the reactor angle ($\sin^2\theta_{13}$)—while the sign and the coefficient ℓ are fixed observable by observable ($1 + \varepsilon/9$ for $\sin\theta_C$, $1 - 2\varepsilon/3$ for $\sin^2\theta_{13}$; derivations in the same paper).

[10] The individual quark magnitudes are treated conditionally in Paper II.

The parity symmetry $V(\phi) = V(-\phi)$ that gave $V_{ub} \approx 0$ also addresses another problem—the strong CP puzzle. $\phi \to -\phi$ means $\sigma \to 1-\sigma$, exchanging occupied ($n = 1$) and empty ($n = 0$)—for fermions this is particle–antiparticle exchange. The Lagrangian inherits this symmetry:

$$\theta_{\text{QCD}} = 0 \qquad (\text{experiment: } |\theta| < 10^{-10}). \tag{24}$$

The observable strong-CP angle is $\bar{\theta} = \theta_{\text{QCD}} + \arg\det M_q$. V-parity forbids the bare $G\tilde{G}$ term—the bare $\theta_{\text{QCD}} = 0$ follows from this structure. That the induced quark mass operator is kept real, giving $\arg\det M_q = 0$, and hence the full $\bar{\theta} = 0$, is given together with its operator-level proof in Paper II. If this holds, the strong CP problem is resolved structurally, without the Peccei–Quinn mechanism or an axion. Yet CP violation *is* observed in the CKM sector ($J = 3.06\times10^{-5}$). This is consistent with the bare $\theta_{\text{QCD}} = 0$: the CKM phase arises at higher order in ε ($J = \pi\varepsilon^5/N_{\text{flags}}$—a theorem under the readout specified in the same paper), while the bare strong-CP angle remains exactly zero.

The consequences of V-parity are summarised:

V-parity consequence	Prediction	Experiment
V_{ub} (tree)	0 (first non-zero term $2\varepsilon^4$, Paper II)	$\lvert V_{ub}\rvert = 0.004$ (generated at higher order)
Bare θ_{QCD}	Exactly zero (structural)	$\bar{\theta} < 10^{-10}$
CKM CP violation	$J = \pi\varepsilon^5/N_{\text{flags}}$ (theorem under the Paper II readout)	3.06×10^{-5} (expt 3.08×10^{-5})

8.4 Neutrinos

§8.1–8.3 derived charged-fermion mass ratios and mixing angles. What remains are the neutrinos.

Mass predictions. Neutrinos are colour singlets with $c = 1$, coupling to $V = -H$—the same potential as charged leptons. The mass ratio R depends only on the well shape (IPR and virial ratios), not on particle species. The same well gives the same R: $R_\nu = R_{\text{lepton}} = 1.529$—this is the input-free prediction of this section. Conditional on the normal-ordering branch and the measured oscillation splittings [28] ($\Delta m^2_{21} = 7.537 \times 10^{-5}$ eV^2, $\Delta m^2_{31} = 2.521 \times 10^{-3}$ eV^2), this ratio fixes the absolute masses[11]:

Quantity	Prediction	Status
m_1	0.32 ± 0.03 meV	Unmeasured
m_2	8.69 meV	Consistent
m_3	50.2 meV	Consistent
$\sum m_i$	59.2 ± 0.3 meV	Below Planck (< 120)

[11]The ±0.03 meV uncertainty propagates from oscillation data input; the theory has no free parameters. Normal ordering is an input condition in this section; its independent derivation is given as a theorem in Paper II (a consequence of $R_\nu > 1$ and $(m_3/m_2)^2 = 169/5 > 1$). What the flag-Laplacian spectrum fixes exactly is $m_3/m_2 = |\text{PG}|/\sqrt{N{+}2} = 13/\sqrt{5}$, i.e. $(m_3/m_2)^2 = 169/5$. The mass-squared-difference ratio is given by the master relation $\Delta m^2_{31}/\Delta m^2_{21} = (169/5 - t^2)/(1 - t^2) = 33.842\ldots$ (displayed as 33.84), obtained by eliminating $t = m_1/m_2$ using $R_\nu = R_{\text{lepton}}$ (Paper II). With non-zero m_1, 169/5 is not the exact value of the mass-squared-difference ratio.

This prediction is falsifiable: the conditional calculation of this section does not itself test the mass ordering, but $\sum m_i > 70$ meV creates tension. Since $\Sigma m_i = 59.2$ meV lies close to the general normal-ordering minimum (58.89 meV at $m_1 \to 0$), a CMB-S4 [27] ($\sigma \sim 15$ meV) detection near this value would confirm the normal ordering but not distinguish this framework from the general minimal scenario; the sharp, framework-specific neutrino prediction is the ratio $\Delta m^2_{31}/\Delta m^2_{21} = 33.8$, testable by JUNO in 2026–27.

Mixing angles. In §8.3, CKM mixing angles were obtained from $\varepsilon = 0.219$—small values ($\sin\theta_C \approx 0.22$). Yet PMNS mixing angles are large ($\sin^2\theta_{12} \approx 1/3$). Why does the same $V = -H$ produce both small and large mixing?

The answer lies in colour charge. Quarks carry colour and reside in the deep well $V = -3H$, which places their mixing in the perturbative regime of small mode overlap—this is why quark mixing is small, and its expansion parameter is the generation-sector marginality ε (§3). Neutrinos are colour-neutral; instead, the geometric allocation of gauge directions on $\mathrm{PG}(2,3)$ directly gives values corresponding to the mixing angles. The solar angle is the fraction of one complete electroweak line ($N{+}1 = 4$ points) in the 13 directions of PG:

$$\sin^2\theta_{12} = \frac{N+1}{|\mathrm{PG}|} = \frac{4}{13} = 0.3077 \qquad \text{(experiment: 0.3088, 0.4\%).} \tag{25}$$

The atmospheric angle is maximal (μ–τ symmetry) plus a PG-scale NLO correction ($k = |\mathrm{PG}| = 13$):

$$\sin^2\theta_{23} = \frac{7}{13}\left(1 + \frac{\varepsilon}{13}\right) = 0.5475 \qquad \text{(second octant).} \tag{26}$$

The sign ($> 1/2$) follows from τ being heavier, predicting the second octant. In NuFIT 6.1 [28] the global best fit has moved to the first octant ($\sin^2\theta_{23} = 0.470$), but the octant is unresolved and the prediction 0.5475 lies within the 3σ range (0.432–0.587); the second-octant test is left to DUNE / Hyper-Kamiokande. The deviation from the second-octant value 0.561 is 2.4% (about 1σ, against the $\sim 2.5\%$ experimental precision). Unless stated otherwise, all NuFIT 6.1 values quoted in this paper are the without-SK-atm, normal-ordering set, consistent with the Δm^2 inputs used above. The reactor angle is $\sin^2\theta_{13} = 1/(N \cdot |\mathrm{PG}|) = 1/39$ with NLO correction ($k = N = 3$):

$$\sin^2\theta_{13} = \frac{1}{39}\left(1 - \frac{2\varepsilon}{3}\right) = 0.0219 \qquad \text{(expt: 0.02249, 2.6\%).} \tag{27}$$

The three PMNS mixing-angle predictions are summarised below:

Angle	LO	NLO prediction	Experiment	Accuracy
$\sin^2\theta_{12}$	$4/13 = 0.308$	—	0.3088	0.4%
$\sin^2\theta_{23}$	$7/13 = 0.538$	0.5475	0.561	2.4%
$\sin^2\theta_{13}$	$1/39 = 0.026$	0.0219	0.02249	2.6%

Consistency check: $\sin^2\theta_{23} = \sin^2\theta_W + \sin^2\theta_{12}$ ($7/13 = 3/13 + 4/13$). The atmospheric mixing angle exhausts the combined fractions of the pure weak (N points) and complete electroweak line ($N{+}1$ points) in $\mathrm{PG}(2,3)$.

Dirac particles. Are neutrinos Dirac or Majorana? This framework gives a clear prediction. Every fermionic operator entering the construction is a function of the number operator $\hat{n}$—the sufficient statistic of the Bernoulli family—and is therefore invariant under the phase rotation $a \to e^{i\theta}a$: the anomaly-free combination $\mathrm{U}(1)_{B-L}$ is exact, and the Majorana bilinear $aa+a^\dagger a^\dagger$ ($\Delta(B-L)=2$) lies outside this operator algebra. V-parity then pairs the distinct states ν and ν^c ($L=\pm 1$) into a Dirac fermion (operator-level proof: the Dirac-selection theorem of Paper II [20]). Therefore neutrinoless double beta decay is exactly zero:

$$0\nu\beta\beta = 0 \qquad \text{(nEXO [29], testable} \sim\text{2030)} \tag{28}$$

With this, fermion mass ratios, mixing angles, and neutrino properties have been obtained from the structure of $V=-H$ and $\mathrm{PG}(2,\mathbb{F}_3)$ (the absolute masses conditional on normal ordering and oscillation data). Where the Standard Model treats these as independent free parameters, this framework treats them all as different aspects of the algebraic expansion of a single equation $V=-H$.

9 Connection to Spacetime

All calculations in §2–8 were performed in internal space (the ϕ coordinate). Neither space nor time appeared. How, then, does this zero-dimensional theory connect to four-dimensional spacetime? This chapter answers in three steps: the dimension and signature $(3,1)$ of spacetime (§9.1), the independence of internal space and spacetime (§9.2), and why the zero-dimensional algebraic values correspond to the four-dimensional experimental values (§9.3).

9.1 Spacetime Dimension and Gravity

Section 4 derived $d_{\text{space}}=3$. But "why four-dimensional spacetime?" is a separate question—it is not obvious that time is one-dimensional. We show that two independent paths give the same $D=4$—both are consistency checks with stated premises (hereafter we distinguish the spatial dimension d from the spacetime dimension $D=d+1$).

The first path comes from gravitational degrees of freedom. A massless spin-2 particle (graviton) in D-dimensional spacetime has $D(D-3)/2$ physical degrees of freedom (following from Poincaré-group representation theory, without assuming any particular D or Einstein's equations). Meanwhile, the $N=3$ bound states of $V=-H$ provide $N-1=2$ independent internal degrees of freedom (since three probabilities sum to one). As a consistency check that takes Poincaré symmetry, a spin-2 field, and gauge reduction as premises, matching the spacetime dynamical degrees of freedom (graviton polarisations) one-to-one with the internal degrees of freedom gives:

$$D(D-3)/2 = N-1 = 2 \quad\Longrightarrow\quad D=4. \tag{29}$$

The second path comes from projective geometry. The projective plane $\mathrm{PG}(2,\mathbb{C}) = \mathbb{C}P^2$ is a real 4-dimensional manifold, and for $N=3$:

$$D = \dim_{\mathbb{R}}(\mathbb{C}P^{N-1}) = 2(N-1) = 4. \tag{30}$$

The two paths are summarised:

Path	Input	Equation	Result
Graviton DOF	Spin-2 polarisations	$D(D-3)/2 = N-1 = 2$	$D=4$
Complex carrier	Complex carrier $\mathbb{C}P^2$	$\dim_{\mathbb{R}}(\mathbb{C}P^{N-1}) = 2(N-1)$	$D=4$

Both are consistent only for $N = 3$.

These two paths are independent—one from field-theoretic representation theory (spin-2 polarisations), the other from pure algebraic geometry (complex carrier $\mathbb{C}P^2$). That both give $D = 4$ is a non-trivial consistency check of $N = 3$.

Even with $D = 4$ fixed, the signature could be $(4, 0)$, $(3, 1)$, or $(2, 2)$. The finite field $\mathbb{F}_3$ (from the primality of $N = 3$) and the complex carrier $\mathbb{C}^3$ (from the unitary realisation) are independent constructions moored to the same $N = 3$—since the characteristics differ, no field homomorphism $\mathbb{F}_3 \to \mathbb{C}$ exists, and no lift via $\mathbb{F}_9$ succeeds either. What links the two is the correspondence between the exponent labels of the Heisenberg–Weyl system ($\mathbb{F}_3$ side) and the representation carrier ($\mathbb{C}^3$ side); what is preserved is the Weyl commutation relations and the MUB structure. $\mathbb{C}^3$ carries a complex structure from the outset. Furthermore, the transfer action $S = \int[\frac{1}{2}\dot{\phi}^2 + V]\,d\tau$ of the internal coordinate ϕ opened by A2 (§2.2) has a single evolution parameter τ, so one of the four real directions is distinguished as the evolution direction. The complex structure and the single evolution direction alone do not uniquely fix the signature. The selection of $(3, 1)$ is fixed as a connection to the determinant form on $M_2(\mathbb{C})_{sa}$ and the structure $\mathrm{SL}_2(\mathbb{C}) \to \mathrm{SO}^+(1, 3)$. The signature is fixed to $(3, 1)$.

In four dimensions, Lovelock's theorem [30] uniquely determines the gravitational field equation: the only differential equation of second order or lower is the Einstein equation

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}.$$

Jacobson [31] showed that the same equation follows from $\delta Q = T_H\, dS$ on a causal horizon. Since the entropy of the present theory is $S = H(P)$, both the Lovelock (geometric) and Jacobson (thermodynamic) paths lead from binary entropy to Einstein gravity.

9.2 Why Internal Space and Spacetime Are Independent

The coordinate $\phi = \mathrm{logit}(\sigma)$ lives in internal space; the coordinate x lives in spacetime. Why are they independent?

The answer lies in the construction of $V = -H$. The potential $V = -H(\sigma(\phi))$ is built as the internal zero-mode of the Bernoulli degree of freedom and contains no explicit spacetime coordinate x. However, the coordinate-independence $\partial V/\partial x = 0$ alone does not exclude a mixed kinetic term. What makes the decomposition hold is the construction in which ϕ (internal) and x (spacetime) are independent degrees of freedom, with generators acting on separate factors and no cross term. Under this, the full Hamiltonian takes the form $H = H_\phi + H_x$, and the Hilbert space decomposes as a tensor product $\mathcal{H}_{\mathrm{gen}}(\phi) \otimes \mathcal{H}_{\mathrm{space}}(x)$.

Since this composite space has dim ≥ 3 ($\mathcal{H}_{\mathrm{gen}}(\phi)$ is 3-dimensional, $\mathcal{H}_{\mathrm{space}}(x)$ is infinite-dimensional), Gleason's theorem [32] uniquely extends the detection probability $\sigma = \langle n \rangle$ of §2 to the standard Born rule—the measurement rule is not an axiom but a consequence of the tensor-product structure.

Two observational facts confirm this. First, all gauge couplings are independent of generation—direct evidence that ϕ and x are not coupled. Second, the 0D theory computes only dimensionless quantities (mass ratios, mixing angles), while dimensionful quantities (absolute masses) require spacetime—precisely the prediction of the tensor-product structure.

The analogy with spin illuminates this structure. Spin algebra is finite-dimensional and independent of the Lagrangian, yet determines 4D physical quantities (magnetic moment, fine structure). Generations have the same structure—the finite-dimensional $V = -H$ corresponds to 4D mass ratios and mixing angles.

9.3 Correspondence Between 0D Values and 4D Experiment

The tensor-product structure explains why 0D algebraic values coincide with tree-level values of the 4D renormalised theory. The 0D values are algebraic constants. In 4D, tree level is the limit of zero loop corrections—i.e. the limit in which spacetime dynamics does not affect internal structure—which is precisely the independence guaranteed by $\mathcal{H}_{\mathrm{int}}(\phi)\otimes\mathcal{H}_{\mathrm{space}}(x)$. Radiative corrections arise from loop momenta in the spacetime factor $\mathcal{H}_{\mathrm{space}}$ and enter as sub-percent corrections to the internal algebraic values (§7: $\sin^2\theta_W = 3/13 + \Delta_{\mathrm{rad}} = 0.23122$). The 0D values are neither RG fixed points nor UV boundary conditions, but the unique algebraically determined point serving as the reference for comparison with 4D theory—running quantities acquire the 4D RG evolution and loop corrections, while quantities such as discrete state counts do not run.

With this, the relationship between internal structure (ϕ) and spacetime (x) is established. $V = -H$ gives internal values corresponding to mass ratios, mixing angles, and coupling constants. The spacetime factor supplies, in addition to the overall energy scale, the 4D RG evolution and loop corrections for running quantities.

10 Results

From the single equation $V = -H$ and the single geometric structure $\mathrm{PG}(2, \mathbb{F}_3)$, we have shown that a mathematical structure matching the structure of the Standard Model is obtained, and that calculations with no adjustable parameters yield values agreeing with twenty physical constants of the Standard Model to 0.0006%–2.6% accuracy. A further six ($|V_{cb}|$, $|V_{ub}|$, δ_{CKM}, δ_{PMNS}, m_3/m_2, $\Delta m^2_{31}/\Delta m^2_{21}$) require readouts built from the specified operator words and the spectral structure of the flag Laplacian, and are derived in Paper II. The full list of results is given below (experimental values from PDG 2026 [22], NuFIT 6.1 [28], and CODATA 2022 [34]). The two largest deviations ($\sin^2\theta_{23}$ at 2.4% and $\sin^2\theta_{13}$ at 2.6%) both lie within about 1σ of the corresponding experimental uncertainties (§8.4).

Table 1. Comparison with Standard Model constants.

Parameter	Prediction	Experiment	Accuracy	§
Gauge couplings				
$1/\alpha_{\rm em}$ (Thomson point)	137.0368	137.036	0.0006%	6.3
α_s ($\overline{\rm MS}$, M_Z)	0.1179	0.1179	0.01%	6.2
$\sin^2\theta_W$ ($\to g_1/g_2$) ($\overline{\rm MS}$, M_Z)	3/13	0.23122	0.2%	6.1
Higgs				
m_H/v ($\to \lambda$)	0.5084	0.5082	0.03%	7
Fermion masses				
$R_{\rm lepton}$ ($\to y_e, y_\mu, y_\tau$)	1.5293	1.5294	0.006%	8.1
R_ν (input-free prediction)	1.529	— (unmeasured)	$R_\nu = R_{\rm lepton}$	8.4
$R_{\rm up}$	1.777	1.770	0.4%	8.2
$R_{\rm down}$	2.273	2.276	0.1%	8.2
CKM mixing				
$\sin\theta_C$	0.2245	0.2243	0.10%	8.3
PMNS mixing				
$\sin^2\theta_{12}$	4/13	0.3088	0.4%	8.4
$\sin^2\theta_{23}$	0.5475	0.561	2.4%	8.4
$\sin^2\theta_{13}$	0.0219	0.02249	2.6%	8.4
Neutrino mass				
m_1	0.32 meV	—	Prediction	8.4
Strong CP				
Bare $\theta_{\rm QCD}$	0 (structural)	$\bar\theta < 10^{-10}$	—	8.3

Table 2. Other results.

Quantity	Derivation	Experiment	§
N (generation number)	$n_{\rm WKB} = 2.781 \to 3$	3	3
d (spatial dimension)	$\{1,3\} \cap \{\geq 3\} \cap \{\geq 3\}$	3	4
D (spacetime dimension)	$D(D-3)/2 = N-1 = 2$	4	9.1
Signature	Connection to $M_2(\mathbb{C})_{sa}$ structure	$(3,1)$	9.1
Gauge group	9+3+1 (Killing–Cartan)	SU(3)×SU(2)×U(1)	5
Spectral gauge verification	Restricted Laplacian products 9+4=13	—	Paper II
Fourth generation	$E_3 > 0$	Excluded	3.2
Neutrino nature	$\mathrm{U}(1)_{B-L}$ ($\hat n$ structure) + V-parity	—	8.4
m_W/m_Z (leading)	$\sqrt{10/13} = 0.8771$	0.8813 (0.5%)	6.1; 7
Q (Koide)	1.499985	1.50001	8.1
$\sin^2\theta_W$ ($3/13 + \alpha\, 135/2197$)	0.23122	0.23122 ($< 0.01\%$)	7

Table 3. Testable predictions.

Prediction	Value	Verification experiment	§
$\Delta m^2_{31}/\Delta m^2_{21}$	master relation (leading $m_3/m_2 = 13/\sqrt{5}$, non-zero m_1 included) = 33.84	JUNO [26] (2026–27)	Paper II
θ_{23} octant	Second ($> 45°$)	DUNE / HK (∼2030)	8.4
$\delta_{\rm PMNS}$	197.1°	DUNE / HK (∼2030)	Paper II
Σm_ν	59.2 meV (≈ NO min. 58.9)	CMB-S4 [27] (∼2030)	8.4
$0\nu\beta\beta$	Exactly zero	nEXO [29] (∼2030)	8.4
Koide deviation δ	-1.5×10^{-5}	Belle II [33] (2027–30)	8.1
BSM particles	None	LHC / FCC	3; 5.3

The Standard Model treats these physical quantities as independent parameters to be measured and input, but the derivation in this paper has no free parameters. The numerical values are based on the semiclassical action count n_{WKB} and finite-grid spectral computation, but the agreement across multiple independent quantities strongly supports the validity of this construction. All numerical results of this paper (including every value in Tables 1–3 and the interval certifications of §3) can be independently reproduced with the verification-code suite provided as supplementary material.

11 Conclusion

This paper has shown that structures and numerical values normally treated as independent inputs in the Standard Model correspond to a single mathematical structure fixed by the three axioms of existence, and are reproduced with no adjustable parameters under that correspondence. The particle content, gauge structure, and more than twenty physical quantities are different aspects of a single equation

$$V = -H(\sigma(\phi))$$

and are not independent. This suggests that the fundamental laws of matter may be described as a necessity from this equation derived from the axioms of existence.

Paper II [20] treats the derivation of the remaining six parameters, the spectral proof of the gauge decomposition 9+3+1, and a more detailed account of the uniqueness of the construction used in this paper. Subsequent papers extend this framework to gravity, the Universe, and the unification of forces.

To be, or not to be — that is the equation.

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